Ahlam Zahran

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Education

University of Petra — Amman, Jordan
B.Sc. in Artificial Intelligence & Data Science ***GPA: 3.59/4.00*** — *Sep 2022 – Jun 2026*
Graduation project: *NeuroSeekers (prototype of Neurender)*

German Jordanian University — Amman, Jordan
Biomedical Engineering (transferred) — *Sep 2019 – Jun 2022*

Experience & Projects

Neurender — Amman, Jordan
Founder & CEO — *2026 – Present*

- Founded Neurender to take the NeuroSeekers prototype from a measured research system to a product: AI that rebuilds the image a person is seeing from their fMRI brain scans, with every result scored, bounded and traced.
- Extending the system to new people without retraining (cross-person decoding; early results reproducible from saved data), toward validation on standard 3-Tesla hospital scanners and a verification standard for brain-decoding results.
- Intended uses: a communication channel for people who are awake but cannot speak or move; longer term, after rigorous validation, forensic research including deception detection.

NeuroSeekers (prototype of Neurender) — University of Petra
AI Engineer & Full-Stack Developer — *2025 – 2026*

- Rebuilt the Takagi et al. (CVPR 2023) fMRI-to-image reconstruction pipeline with text-guided semantic decoding, reaching **82.1%** mean two-way identification accuracy on 982 held-out images (chance 50%).
- Trained subject-specific decoders on the Natural Scenes Dataset (NSD): ROI-wise encoding with ridge regression and latent-space decoding; diffusion-based image generation with a blind candidate selector.
- Built the verification framework: measured noise ceiling, pre-registered tests, documented failures, and a provenance record for every result.
- Built a full-stack web application (Django backend, React + Bootstrap frontend) with a live demo, real-time result display and visual analysis tools.
- Designed workflows for cognitive-response monitoring and forensic research, including deception detection.
- Presented the project at **SOFEX Jordan C8 Cybersecurity Exhibition 2025**, representing the University of Petra under the patronage of HRH Crown Prince Al Hussein bin Abdullah II; strong interest from cybersecurity, AI and forensics experts; invited back to present at SOFEX Aqaba 2026.

Chair – IEEE Computational Intelligence Society — University of Petra
Student Chapter Chair — *2024*

- Led and coordinated technical events, workshops, and outreach initiatives to advance computational intelligence on campus.
- Strengthened leadership, organizational, and cross-functional collaboration skills across student and faculty teams.

IoT Instructor — University of Petra
Workshop Instructor — *2024*

- Conducted hands-on workshops on Arduino-based IoT projects for university students.
- Guided students in building and deploying autonomous robotic systems from prototype to demonstration.

AWS DeepRacer AI Model Developer — AWS Global Competition
Top 500 Globally — *2023*

- Designed and trained autonomous racing models using Proximal Policy Optimization (PPO) via AWS SageMaker.
- Ranked among the top 500 participants globally out of thousands of competitors.

Hackathon & Competition Projects

2nd Place – World Robotics Championship Competition (WRCC), Open Category 2026

- Awarded **Second Place** in the international WRCC Open Category for designing and developing an AI-powered autonomous robotic system combining robotics, computer vision, and intelligent decision support.
- Competed against international university teams.

Peer-to-Peer Energy Trading Platform | *WEFE Hackathon – 1st Place Winner* 2024

- Conceived and pitched a business idea for an AI-driven peer-to-peer energy trading platform proposing Reinforcement Learning and a Greedy Algorithm to optimize renewable energy distribution.
- Designed the full UI/UX prototype in Figma; won first place among competing teams on concept, feasibility, and presentation.

Skills

Programming Languages: Python, Java, SQL

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Computer Vision, NLP, Generative AI (diffusion models), Reinforcement Learning

Neuroimaging & Data: fMRI analysis (NSD), ridge-regression decoding, Data Mining, Big Data Analytics, Data Warehousing, Power BI

Cloud & DevOps: AWS SageMaker, S3, RAG (Retrieval-Augmented Generation), Git/GitHub

IoT & Robotics: Arduino, ESP32, Autonomous Robotics, ROS2

Web & Tools: Django, React, Bootstrap, Figma

Certifications & Training

DAAD Business Training | Germany **(2024)** – Competitive international program on strategy, entrepreneurship, and global market trends.

Generative AI with AWS | Udacity **(2024)** – Built an AI chatbot using supervised/unsupervised learning; applied AWS SageMaker and RAG.

Erasmus+ RL4Eng | Bonn-Rhein-Sieg University, Germany **(2023)** – FPGA programming, VHDL, hardware design, and international collaboration.

Multi-Robot Drone Workshop | Sager Company **(2023)** – Built a drone; applied in logistics, security, and monitoring scenarios.

Marketing Workshop | Jaguar & Land Rover **(2023)** – Insights into luxury brand marketing strategy and global consumer trends.

Languages

Arabic: Native **English:** Fluent **German:** Intermediate